New Sobolev Embedding Theorem

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If the domains of functions is R^N then embedding theorem follows even if on bounded domains without extension operators.

The photograph shows an example of a function with low regularity that does not belong to a Sobolev space imposing high differentiability near the origin. Another example is

$$x \mapsto |x|.$$

This function belongs to the Sobolev space

$$W^{1,2}(\Omega)$$

on any bounded open interval Ω, which may contain the origin. However, its weak derivative from the right and its weak derivative from the left do not match at the origin, and differentiating it weakly once more gives a delta distribution. The delta distribution is not an L^p function for any

$$1 \leq p \leq \infty.$$

On the other hand, if

$$s \in R, \qquad s < -\frac{N}{2},$$

then the delta distribution is an element of $W^{s,2}$. Nevertheless, the delta distribution does not belong to Sobolev spaces of nonnegative integer order. Therefore,

$$x \mapsto |x|$$

does not belong to $W^{2,2}(\Omega)$.

The Fourier transform (and inverse Fourier transform) converts partial differential operators into multiplication operators given by polynomials. Let $\mathcal{S}$ denote the Schwartz space, i.e. the space of smooth functions that, together with all their derivatives, tend to zero at infinity faster than any inverse polynomial. For $u \in \mathcal{S}$, define

$$\mathcal{F}[u](\xi) := \frac{1}{(2\pi)^{N/2}} \int_{R^N} e^{-i\xi\cdot x} u(x)\, dx,$$

and

$$\mathcal{F}^{-1}[u](\xi) := \frac{1}{(2\pi)^{N/2}} \int_{R^N} e^{i\xi\cdot x} u(x)\, dx.$$

These are continuous linear bijections on $\mathcal{S}$, and integration by parts gives

$$\mathcal{F}[\partial^\alpha u](\xi) = (i\xi)^\alpha \mathcal{F}[u](\xi).$$

Moreover, $\mathcal{S} \subset L^2$ is dense in L^2, and the extended Fourier transform is an isometry on L^2. The identity

$$\partial^\alpha u(x) = \mathcal{F}^{-1}\left[(i\xi)^\alpha \mathcal{F}[u](\xi)\right](x)$$

provides the basic idea for defining derivatives of fractional order.

The inverse Fourier transform really is the inverse of the Fourier transform. In mathematics, before it has been proved that the inverse Fourier transform is indeed the inverse mapping of the Fourier transform, it is sometimes called the conjugate Fourier transform.

Proposition 1 *Let $k, m, N \geq 0$ be nonnegative integers satisfying*

$$m > k + \frac{N}{2}.$$

Then

$$W^{m,2}(R^N) \subset B^k(R^N),$$

and this embedding is continuous. Moreover, for every open set

$$\Omega \subset R^N,$$

we have

$$W^{m,2}(\Omega) \subset C^k(\Omega).$$

For the latter embedding, if we consider functions whose domain is R^N, continuity of the embedding also holds.

Meaning

A Sobolev space consists of equivalence classes of weakly differentiable functions. Weak regularity is measured by the integrability of weak derivatives, or equivalently by the norm appearing in the definition of the Sobolev space. Although the differentiability is understood in a weak sense, a function whose regularity is too poor does not belong to the Sobolev space.

If a sufficient degree of weak regularity is assumed, then there exists a representative that is smooth in the ordinary sense to the prescribed order, and whose partial derivatives are bounded. Roughly speaking, if a function in a Sobolev space has sufficiently much weak regularity, then its values can be modified on a set of measure zero so that it becomes smooth in the ordinary sense.

The B^k-norm is the sum of the suprema of the absolute values of all partial derivatives of order at most k (or, equivalently, one may use the maximum of these suprema).

Continuity of the embedding means that there exists a constant

$$0 < c < \infty$$

independent of u such that

$$\|u\|_{B^k} \leq c\|u\|_{W^{m,2}}.$$

If c were allowed to depend on u, such an inequality could always be obtained, and this would cause serious difficulties in the analysis of partial differential equations: inequalities cease to be useful when the various multiplicative constants appearing in them depend on the unknown function.

Since

$$\|u - v\|_{B^k} \leq c\|u - v\|_{W^{m,2}},$$

if u and v are close in the topology of the Sobolev space, they are also close in the topology of the space of smooth functions. Indeed, for every $\varepsilon > 0$, taking

$$\delta = \frac{\varepsilon}{c+1},$$

we have

$$\|u - v\|_{W^{m,2}} < \delta \quad \Longrightarrow \quad \|u - v\|_{B^k} < \varepsilon.$$

Thus the embedding map

$$W^{m,p} \ni u \longmapsto u \in B^k,\ C^k$$

is continuous.

Lemma 1 *There exist constants $C_m, c_m > 0$ such that*

$$c_m(1 + |\xi|^2)^m \leq \sum_{|\alpha| \leq m} |\xi^\alpha|^2 \leq C_m(1 + |\xi|^2)^m.$$

The upper bound follows from the multinomial theorem. For the lower bound, it suffices to take c_m to be the reciprocal of the largest multinomial coefficient.

[Proof of the Proposition] Let

$$u \in \mathcal{S}.$$

By the Fourier inversion formula and the Cauchy–Schwarz inequality,

$$|\partial^\alpha u(x)| \leq c_N \int_{R^N} (1 + |\xi|^2)^{k/2 - m/2}(1 + |\xi|^2)^{m/2} |\mathcal{F}[u](\xi)|\, d\xi.$$

Hence

$$|\partial^\alpha u(x)| \leq c_N \int_{R^N} (1 + |\xi|^2)^{-N/4}(1 + |\xi|^2)^{m/2} |\mathcal{F}[u](\xi)|\, d\xi \leq c'_N \|u\|_{W^{m,2}}.$$

Therefore,

$$\|u\|_{B^k} \le c'_N \|u\|_{W^{m,2}}.$$

Now let

$$u \in W^{m,2}(R^N).$$

Since $\mathcal{S}$ is dense in $W^{m,2}(R^N)$, choose

$$\{u_n\} \subset \mathcal{S}$$

such that

$$u_n \to u \qquad \text{in}\, W^{m,2}(R^N).$$

By the inequality above, $\{u_n\}$ is also a Cauchy sequence in B^k. Since B^k is complete, there exists a uniform limit

$$u' \in B^k.$$

The uniform convergence shows that $u = u'$ almost everywhere. Indeed, for every $\varepsilon > 0$, there exists $\nu \in N$ such that, whenever $n > \nu$,

$$\sup_{x \in R^N} |u_n(x) - u'(x)| \le \varepsilon.$$

Consequently, for every bounded Jordan measurable set A,

$$\sup_{x \in A} |u_n(x) - u'(x)| \le \varepsilon,$$

and therefore

$$\|u_n - u'\|_{L^2(A)} \le |A|^{1/2}\varepsilon.$$

Since $u_n \to u$ in $L^2(A)$, it follows that $u = u'$ almost everywhere on A. Since A is arbitrary,

$$u = u'$$

almost everywhere on R^N. Moreover,

$$\|u_n\|_{B^k} \to \|u\|_{B^k}, \qquad \|u_n\|_{W^{m,2}} \to \|u\|_{W^{m,2}}.$$

Thus the embedding inequality follows. (The argument in this paragraph is my own.)

Finally, let

$$\omega$$

be open, and choose a smooth compactly supported function φ such that

$$\varphi \equiv 1 \qquad \text{on}\, \omega.$$

Then

$$\varphi u \in W^{m,2}(R^N),$$

and hence

$$u \in B^k(\omega).$$

Since $\omega\Omega$ is arbitrary, we obtain

$$u \in C^k(\Omega),$$

which proves the stated inclusion.

If the functions under consideration are defined on all of R^N, then, for a suitable constant c'_N, one may choose a smooth compactly supported function φ satisfying

$$\varphi \equiv 1 \qquad \text{on}\, \Omega.$$

The embedding above can then be applied to φu. Since φ may be chosen as needed, continuity of the embedding follows as well.

1 References

Robert A. Adams and John J. F. Fournier, Sobolev Spaces, Academic Press, 2003; S.Miyajima, Basic of Sobolev Spaces, Japan Review Company, 2006, ISBN: 9784320018280.